

AHMAD RIPAI

🏛️ P.h.D. Student in Theoretical Physics

📍 Theoretical Physics Division, Department of Physics, IPB University

Theoretical Geophysical Fluid Dynamics, Research Center for Climate and Atmosphere, BRIN

RESEARCH INTEREST

Primary research interests lie in the theoretical physics of complex systems, with a particular emphasis on nonlinear wave phenomena. Current work in theoretical geophysical fluid dynamics focuses on nonlinear atmospheric wave dynamics and land–sea breeze circulation in the Maritime Continent. Previous research spans nonlinear optics, biophysics, and quantum physics, including studies on solitons, rogue waves, Airy beams, nonlinear DNA dynamics, and Bose–Einstein condensates.

Keywords: theoretical physics, nonlinear waves, solitons, geophysical fluid dynamics.

PERSONAL

Year of Birth : 1996
Place of Birth : Muara Botung, North Sumatra Province, Indonesia.
Nationality : Indonesia.
Pronoun : He/Him/His.
Language : Indonesian (Native), English (Intermediate).

EDUCATION

Doctor of Philosophy (Ph.D.) in Physics *Now-2026*
on going, GPA: -/-

- Student Number (NIM)—G7601241001
- Research topics—Nonlinear atmospheric wave dynamics and land–sea breeze circulation
- Institution : Department of Physics, IPB University.
- Supervisor : Prof. Dr. Husin Alatas
- Co-Supervisor : Dr. Albertus Sulaiman
- Support : BRIN - Degree by Research (Full).

Master of Science (M. Sc.) in Physics *August 2019 - May 2020*
Cumlaude, GPA: 3.89/4.00.

- Thesis—Split-step Fourier method for photorefractive solitons
- Institution : Department of Physics, Universitas Andalas.
- Supervisor : Dr. Zulfi Abdullah.
- Co-Supervisor : Dr. Mahdhivan Syafwan (Mathematics) and Dr. Wahyu Hidayat (Physics, ITB).
- Support : Universitas Andalas - Fast Track Program (Full).

Bachelor of Science (B. Sc.) in Physics *August 2015 - May 2019*
Cumlaude, GPA: 3.67/4.00.

- Thesis—Analytical soliton solutions of the Burgers equation
- Institution : Department of Physics, Universitas Andalas.
- Supervisor : Dr. Zulfi Abdullah.
- Co-Supervisor : Dr. Mahdhivan Syafwan (Mathematics).
- Support : Government of Indonesia - Bidikmisi (Full).

RESEARCH EXPERIENCE

Research Assistant

2024 - Now

Research Assistant Program, National Research and Innovation Agency (BRIN)

Institution : Research Center for Climate and Atmosphere, BRIN.
Group : Theoretical Geophysical Fluid Dynamics.
Supervisor : Dr. Albertus Sulaiman.
Project : Nonlinear atmospheric wave dynamics and land-sea breeze circulation.

Research Assistant

2023 - 2023

Matching Grant between Universitas Andalas and Universiti Malaysia Perlis

Institution : Department of Physics, Universitas Andalas.
Supervisor : Dr. Zulfi Abdullah.
Grant : IDR 46,000,000.
Project : Quantum Soliton Fluctuations in Quantum Reflection of BECs.
Collaborator : Dr. Mohamad Nazri Abdul Halif (Theoretical Physics, IMK, UNIMAP).

Research Fellow

2021 - 2022

Fundamental and collaborative research grants, FMIPA, LPPM, Universitas Andalas

Institution : Department of Physics, Universitas Andalas.
Supervisor : Dr. Zulfi Abdullah.
Grant : IDR 120,000,000.
Project : Mathematical Modelling and Theoretical Aspect of Photorefractive Soliton.
Collaborator : Dr. Aavishkar Katti (Physics, MIT-WPU, India).

Research Assistant

2020 - 2021

Junior lecturer research grants, FMIPA, Universitas Andalas

Institution : Department of Physics, Universitas Andalas.
Supervisor : Trengginas Eka Putra Sutantyo, M.Si.
Grant : IDR 15,000,000.
Project : Nonlinear Dynamics of DNA.

PEER-REVIEW SERVICE

Referee for *Optical and Quantum Electronics* (Springer Nature), *Nonlinear Dynamics* (Springer Nature), *Scientific Reports* (Nature Portfolio).

BOOKS

Soliton dalam Medium Fotorefraktif: Pengantar, Teori, dan Komputasinya

Solitons in Photorefractive Media: Introduction, Theory, and Computation

Language: Indonesian (Bahasa)

Status: Under Publication

Preprint: Available on ResearchGate (Ahmad Ripai)

PUBLICATION AND PREPRINT

Selected Publications (Peer-Reviewed Journals and Proceedings)

10. Ripai, A., Musyayyadah, H. A., Abdullah, Z., & Katti, A. (2025). Solitons, breathers, and rogue waves on photorefractive backgrounds: exact solutions and NLSE-NLSE transformations. *The European Physical Journal Plus*, 140(1), 65.
9. Ripai, A., Setiawan, A., Laksanawati, W. D., Kamaruddin, A., Darmawan, N., Iskandar, J., & Irzaman, I. (2025). The effect of ionic radius carbon dot on Ti⁴⁺ in lattice parameters of Ba_{0.2}Sr_{0.8}TiO₃ thin films. In *BIO Web of Conferences* (Vol. 171, p. 02008). EDP Sciences..

8. Musyayyadah, H. A., Halif, M. N. A., Ripai, A., & Abdullah, Z. (2024). Optimizing quantum reflection of Bose–Einstein condensates with angle of incidence. *Physica Scripta*, 100(1), 015414.
7. Ripai, A., Abdullah, Z., & Musyayyadah, H. A. (2024). Solitonic characteristics of optical Airy beams nonlinear propagation in biased centrosymmetric photorefractive medium. *Optics Communications*, 570, 130932.
6. Abdullah, Z., Ripai, A., Musyayyadah, H. A., Sutantyo, T. E. P., Syafwan, M., Hidayat, W., ... & Katti, A. (2024). Temporal behavior of diffusion-trapped Airy beams in photorefractive media. *Optics Communications*, 550, 129930.
5. Ripai, A., Sulaiman, A., Alatas, H., Ogawa, M., Kozan, O., Noersomadi, ... & Yamanaka, M. D. (2023, November). On the Atmospheric Solitary Waves Propagation Over Bengkalis Island. In *International Conference on Radioscience, Equatorial Atmospheric Science and Environment* (pp. 475-484). Singapore: Springer Nature Singapore.
4. Abdullah, Z., Ripai, A., Syafwan, M., & Hidayat, W. (2023). Traveling wave solutions for explicit-time nonlinear photorefractive dynamics equation. *Nonlinear Dynamics*, 111(17), 16515-16526.
3. Abdullah, Z., Ripai, A., Musyayyadah, H. A., Sutantyo, T. E. P., Syafwan, M., Hidayat, W., ... & Halif, M. N. A. (2023). Temporal behavior of bright and dark spatial solitons in photorefractive crystals having both the linear and quadratic electro-optic effects based on low amplitude approximations. *Optik*, 284, 170871.
2. Sutantyo, T. E. P., Ripai, A., Abdullah, Z., Hidayat, W., & Zen, F. P. (2022). Soliton-like solution on the dynamics of modified Peyrard-Bishop DNA model in the thermostat as a bio-fluid. *Emerg. Sci. J.*, 6(4).
1. Ripai, A., Abdullah, Z., Syafwan, M., & Hidayat, W. (2021, April). Application of the split-step Fourier method in investigating a bright soliton solution in a photorefractive crystal. In *AIP Conference Proceedings* (Vol. 2331, No. 1, p. 030023). AIP Publishing LLC.

Research Profiles



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CONTRIBUTED TALK

Oral Presentation

- *Harnessing Physics-Informed Neural Networks for Dynamical Modeling of MJO Propagation across the Maritime Continent*, China-Indonesia Ocean and Atmosphere Training Workshop, November 16-17th, 2025.
- *Climate and Atmospheric Dynamics over The Maritime Continent*, The 3rd International Conference on Radioscience, Equatorial Atmospheric Science and Environment (INCREASE), November 21-22th, 2023.
- *Optical Spatial Soliton in Photorefractive Crystals*, National Conference Cluster and Downstream Advance Research (KN-KHRB) VII 2021, Universitas Andalas, December 13-19th, 2021.
- *The Property Of Photorefractive Soliton Based On Separate Review Of Linear And Quadratic Electro-Optic Effects*, National Conference Cluster and Downstream Advance Research (KN-KHRB) VII 2021, Universitas Andalas, December 13-19th, 2021.
- *Application of the Split-Step Fourier Method in Investigating a Bright Soliton Solution in a Photorefractive Crystal*, National Conference Cluster and Downstream Advance Research (KN-KHRB) VI 2020 Universitas Andalas, December 1st-4th, 2020.

- *Effect of Ansatz on Soliton Propagation Pattern in Photorefractive Crystals*, The 3rd International Conference on Research and Learning Physics (ICRLP), State University of Padang, September 3rd-4th, 2020.
- *Application of the Split-Step Fourier Method in Investigating a Bright Soliton Solution in a Photorefractive Crystal*, The 2nd Science and Mathematics International Conference (SMIC), State University of Jakarta, August 8-9th, 2020.
- *Benchmarking of the Split-Step Fourier Method in Solving a Soliton Propagation Equation in a Non-linear Optical Medium*, Seminar of National Physics, State University of Jakarta, June 20th, 2020.

AWARDS

Exam maker for National Physics Olympiad, Physics Festival, Universitas Andalas	2021
The best graduan of Department of Physics, Universitas Andalas	2019
Finalist of Physics Competition, MIPA Expo, Universitas Riau	2019
Semifinalist of Physics Competition, MIPA Expo, Universitas Riau	2019
Judge of National Physics Olympiad, Physics Festival, Universitas Andalas	2018
Contingent of National Physics Olympiad, ON-MIPA PT, X Region Kopertis, Pekanbaru	2018
Contingent of National Physics Olympiad, ON-MIPA PT, X Region Kopertis, Pekanbaru	2017
Runner up Physics Olympiad, POIF, Universitas Bengkulu	2017
* * * National level awards	

TEACHING EXPERIENCE

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- Tutor, National Physics Olympiad, Senior High School, West Sumatra Province, Indonesia, (2023).
 - Teaching Assistant, Mathematical Physics 1 & 2, Soliton Theory, Electrodynamics, Modern Optics, Classical Mechanics, and Quantum Mechanics, Universitas Andalas, Indonesia (2017-2022).
 - Laboratory Assistant, Computational Physics Laboratory, Univeritas Andalas, Indonesia (Second Half of 2017 and First Half of 2018).
 - Tutor, National Olympiad in physics, college, Universitas Andalas, Indonesia, (2018-2019).

SKILLS

Analytics	Mathematical modelling and analytics, Mathematical physics
Computations	Computational physics and mathematics, numerical method
Programming Languages	Python, MATLAB, Wolfram Mathematica
Python Packages	Pandas, Matplotlib, Numpy, Scipy, Jupyter
Software	LaTeX, Microsoft Office or 365, Mathematica, COMSOL

REFERENCE

Prof. Husin Alatas	Department of Physics, IPB University alatas@apps.ipb.ac.id
Dr. Albertus Sulaiman	Research Center for Climate and Atmosphere, BRIN albe002@brin.go.id